

Final report UNITA STARTING GRANT

Date	Sep. 28, 2026
Research Hub	Digital Transition
Title of the starting grant call	Green and Secure ICT for the digital society in Anthropocene
Acronym	BE-SAFE – Bicycle SAFETY for Mobility and Active Riding in Smart Cities
Budget allocated (UBI)	€2,500.00 (total budget: €17,500.00)
Project PI	Bertha Santos
Leading university	University of Beira Interior
Other universities involved	Universidad Pública de Navarra (Spain), Università degli Studi di Brescia (Italy), Transilvania University of Braşov (Romania)

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PREAMBULE

The Call for Starting Grants on Research aim to strengthen collaboration between UNITA partners through the implementation of new projects or the consolidation of existing projects.

I. FULL TITLE OF THE PROJECT

BE-SAFE – Bicycle SAFETY for Mobility and Active Riding in Smart Cities

II. PROJECT DESCRIPTION

BE-SAFE (Bicycle SAFETY for Mobility and Active Riding in Smart Cities) was a multidisciplinary UNITA Starting Grant project aimed at improving cyclist safety through the integration of data analytics, behavioural research, sensing technologies, V2X communications, and simulation-based assessment. The project brought together expertise from transportation engineering, psychology, telecommunications, computer science and urban planning to develop an integrated framework for cyclist safety assessment and decision support.

The project addressed cyclist safety from four complementary perspectives: (i) analysis of bicycle accident patterns using GIS and Machine Learning techniques, (ii) identification of cyclist behavioural profiles through large-scale international surveys, (iii) evaluation of sensing technologies and communication architectures for bicycle monitoring and V2X applications, and (iv) simulation of cyclist–vehicle interactions using SUMO-based digital environments.

Major achievements include the analysis of 3,807 bicycle-related accidents, the collection of 740 cyclist questionnaires across four countries, the design of an IoT-based sensing architecture, and the development of simulation scenarios capable of evaluating cyclist safety under different behavioural conditions. These outcomes culminated in the development of an integrated cyclist safety framework and the presentation of project results at the European Transport Conference (ETC 2026).

III. DESCRIPTION OF THE CONSORTIUM

The BE-SAFE consortium involved four UNITA universities bringing complementary expertise to address cyclist safety challenges in smart cities. The project was coordinated by **Universidade da Beira Interior**

(Portugal), which led project management, accident analysis, GIS modelling, Machine Learning applications and framework integration.

Universidad Pública de Navarra (Spain) contributed expertise in IoT, sensing technologies, telecommunications, Smart City platforms and V2X communications, leading the development of sensing and connectivity assessment activities.

Università degli Studi di Brescia (Italy) led simulation activities, contributing expertise in vehicular networking, mobility simulation and behavioural modelling within the SUMO framework.

Transilvania University of Braşov (Romania) coordinated the behavioural analysis activities, including survey design, cyclist profiling and behavioural modelling.

The project also engaged nine non-academic stakeholders, including municipalities, road safety authorities, digital innovation hubs, cycling associations and regional mobility agencies, ensuring strong links between research outcomes and real-world applications.

IV. HIGHLIGHTS

- **Events organised**

Two formal consortium meetings were organised during the project:

Mid-term consortium meeting (Braşov, Romania, March 2026).

Closing consortium meeting (Covilhã, Portugal, September 2026).

Continuous online coordination meetings were also held throughout the project to integrate Tasks 1–4 and prepare future collaborative initiatives.

- **Scientific achievements**

Task 1 – GIS & Machine Learning Risk Analysis

A comprehensive database of bicycle accidents was compiled and analysed using GIS and Machine Learning techniques. The study examined 3,807 bicycle-related accidents occurring between 2017 and 2024 in the Lisbon Metropolitan Area. Spatial clustering analyses, Emerging Hotspot Analysis and predictive modelling identified critical accident hotspots and the main factors influencing injury severity, providing the empirical basis for subsequent simulation activities.

Task 2 – Cyclist Behavioural Analysis

A survey involving 740 cyclists from Portugal, Romania, Italy and Spain was conducted. Statistical analysis identified and validated two consistent cyclist behavioural profiles: Safety-Oriented Cyclists and High-Risk Cyclists. These behavioural profiles became a key input for simulation scenario development and future intervention assessment.

Task 3 – Sensors and V2X Technologies

The project evaluated multiple sensing platforms, communication technologies and IoT architectures for cyclist monitoring. A framework for an ESP32-based sensing platform integrating environmental and motion sensors was conceptually specified. Connectivity analyses demonstrated the applicability of WiFi, LPWAN, LTE and 5G technologies for urban cyclist monitoring and V2X applications.

Task 4 – SUMO-Based Simulation

Cyclist behavioural profiles identified in Task 2 were translated into SUMO simulation parameters and incorporated into a digital mobility environment. Three cyclist behaviour scenarios (Safety, Mixed and Risky) were simulated and assessed using surrogate safety measures (TTC – Time To Collision), PET – Post Encroachment Time and DRAC – Deceleration Rate to Avoid a Crash). Results demonstrated that behavioural differences significantly influence cyclist-related conflict indicators and safety outcomes.

- **Links developed with stakeholders**

The project strengthened collaboration with several municipalities and institutions involved in smart mobility and road safety, including the Municipality of Covilhã, the Municipality of Pamplona, ANSR (Portuguese National Road Safety Authority), ESRI Portugal, Braşov Metropolitan Agency, ACS Kronstadt Cycling, IRIS Digital Innovation Hub, and local and regional mobility organisations. These collaborations enhanced access to datasets, validation

opportunities and future implementation pathways.

- **Difficulties encountered**

The project faced challenges associated with the integration of heterogeneous datasets and methodologies across disciplines. Additional difficulties included limitations associated with accident reporting databases, the need to harmonise behavioural and simulation parameters, and constraints related to the availability of real-world sensing deployments within the project timeframe. However, these challenges were successfully mitigated through continuous collaboration between partners and iterative validation of project outputs.

- **Publications and dissemination**

The project generated a collaborative scientific contribution (article) presented at the European Transport Conference (ETC 2026), consolidating the outcomes of all project tasks within an integrated cyclist safety framework (September 2026, Porto). In addition, project dissemination was supported through the development of the BE-SAFE website (<https://besafe-unita.github.io/besafe/>) and key presentations to stakeholders and academic communities, including:

1. UNITA DT Workshop Starting Grant (October 2026, online): Presentation of the project's research framework, findings and next steps – BE-SAFE Next.
2. UNITA Starting Grant dissemination event (September 2026, Brasov, UniTBV): Presentation of the project's research framework, findings and next steps – BE-SAFE Next.
3. ETC 2026 European Transport Conference (September 2026, Porto): Presentation of the paper "BE-SAFE: Initial developments of a data-driven framework for cyclist safety in smart cities".
4. SPARK International Partnership Meeting (April 2026, Covilhã, UBI): Presentation of BE-SAFE's cycling safety framework to international mobility stakeholders and European partners from eight countries during the Interreg SPARK project meeting.
5. AIMAS 2026 (April 2026, Sicily): Presentation of the project's research framework and ongoing findings at the International Conference on Accessibility and Inland Mobility Advancement Strategies.
6. UNITA DT mid-term webinar SG (March 2026, online): Presentation of the project's research framework and ongoing findings.

- **Extra funding (funds by stakeholders or other sources)**

The participation costs for PI Bertha Santos and UBI TM Jorge Gonçalves in the European Transport Conference (ETC 2026) were funded by the UBI members' individual cost centers in the amount of **€1,207.50**, as the limited budget allocated to UBI (€2,500.00) was insufficient to cover this expense.

QUANTITATIVE DATA			
Category	Please put your answer in this column	Category	Please put your answer in this column
Number of participants in your team	10	Among whom:	
		Top grade researcher (Full professor/Director of research)	2
		Senior Researcher (Senior Researcher/ Associate professor)	2
		Recognised researcher (Researcher/ Assistant professor)	4
		Recognised researcher (Post-Doctoral researcher)	0
		First stage researcher (PhD student or other first stage researcher)	2
Number events organized (scientific meetings)	2	Total number of people attending these meetings	6 (mid-term meeting) 5 (closing meeting)
Number events organized (dissemination meetings/ presentations / websites)	6 presentations 1 project website	Total number of people attending these meetings (if known)	9 team members participation Events participants
Number of publications	1	Among whom, in open access	1
Extra funds (funds by stakeholders or third parties), if applicable	€1,207.50 funded by the UBI members' individual cost centres for ETC 2026 registration costs.		
Number of stakeholders effectively involved and typology of the contribution	2 municipalities (Lisbon, Brasov) and 4 institutions involved in smart mobility and road safety, including the ANSR (Portuguese National Road Safety Authority), ESRI Portugal, ACS Kronstadt Cycling, Portuguese Cycling Federation (UVP-FPC) These collaborations enhanced access to datasets, validation opportunities and future implementation pathways.		

V. KEY ACHIEVEMENTS

During its 12-month duration, BE-SAFE successfully established an integrated multidisciplinary framework combining accident analysis, behavioural research, sensing technologies, V2X communications, and simulation-based safety assessment. The project demonstrated the feasibility of linking historical accident intelligence, behavioural profiling, real-time sensing concepts, and digital simulation into a continuous cyclist safety knowledge ecosystem.

A major achievement was the development of the BE-SAFE Integrated Cyclist Safety Framework, which connects:

Task 1: GIS and Machine Learning risk analysis;

Task 2: Behavioural intelligence and cyclist profiling;

Task 3: Sensing, IoT, and V2X technologies;

Task 4: Digital Twin and simulation capabilities.

The framework (see Figure 1) provides a pathway from descriptive and predictive safety analysis towards prescriptive and adaptive safety management, supporting future decision-making processes for municipalities, road safety agencies, and mobility operators (Cyclist Safety Knowledge Hub).

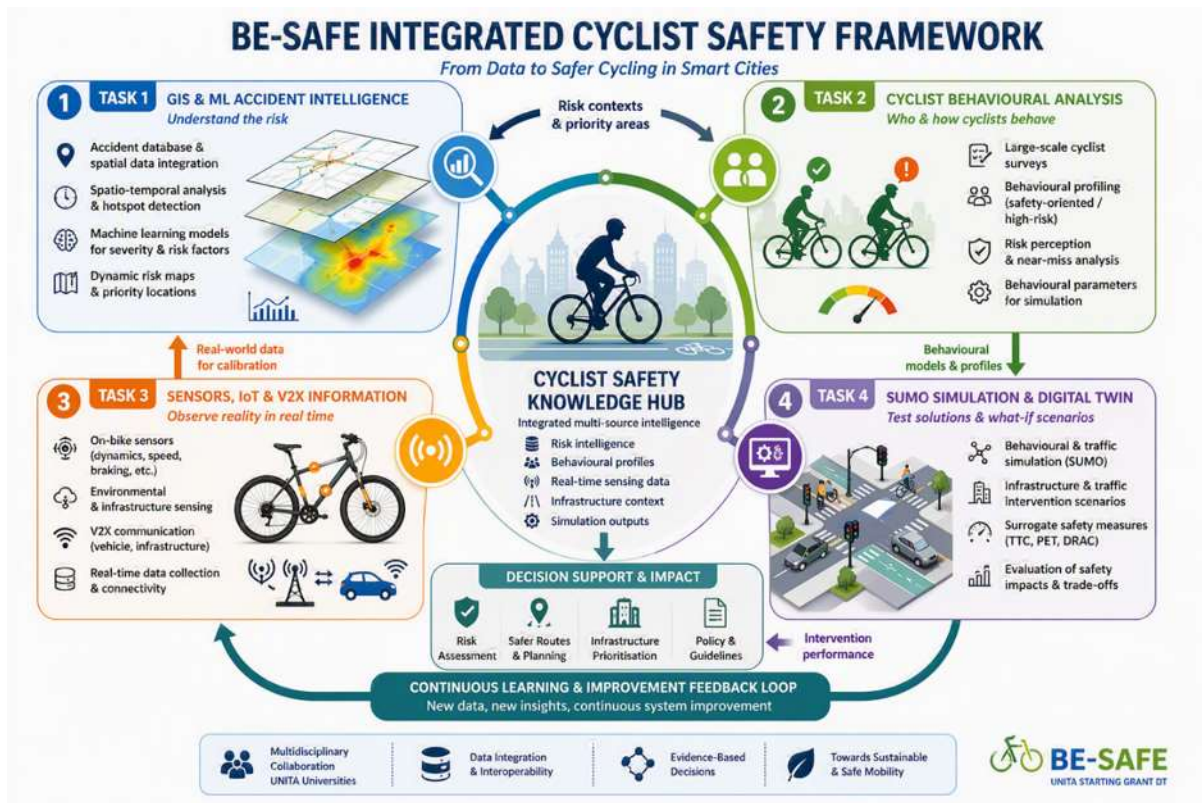


Figure 1 – BE-SAFE Integrated Cyclist Safety Framework. Source: Developed by the authors using Microsoft Copilot (Generative AI), 2026.

A collaborative scientific contribution detailing these advancements—titled "BE-SAFE: Initial Developments of a Data-Driven Framework for Cyclist Safety in Smart Cities" (authored by Bertha Santos, Jorge Gonçalves, Jonathan Sandoval, Francisco Falcone, Renato Lo Cigno, Lorenzo Ghiro, and Ruxandra-Cristina Stanescu)—was accepted and presented at the 54th European Transport Conference (ETC 2026) at the University of Porto, Portugal.

Building on these milestones, the follow-up proposal BE-SAFE Next (Bicycle SAFETY powered by AI and NEXT-gen Communication Toward Cognitive Cities) was submitted to the UNITA Advanced Grants call and officially selected for funding. The expanded initiative incorporates two new academic partners: Université de Pau et des Pays de l'Adour (UPPA) and Haute École Spécialisée de Suisse Occidentale (HES-SO). It also integrates 9 supporting partners representing key regional stakeholders, municipalities, transport authorities, and innovation hubs: [Associazione Italiana per l'Ingegneria del Traffico e dei Trasporti \(AIIT\)](#), [Cluster Smart Cities & Communities Lombardia](#), [Gruppo Community Rotary \(GROC\) – Safety Road for Life](#), [ACS Kronstadt Cycling](#), [Covilhã City Council](#), [ESRI Portugal](#), [Ville de Pau](#), [Brescia Mobilità S.p.A.](#)

VI. PERSPECTIVES

Beyond achieving its scientific objectives, the project created a sustainable research network within the UNITA Alliance and generated a clear roadmap for future development.

A major indicator of the project's success is the continuation of this research through the BE-SAFE-Next (Bicycle SAFETY powered by AI and NEXT-gen Communication Toward Cognitive Cities) initiative, funded under the UNITA Advanced Grant programme. BE-SAFE-Next directly builds upon the results obtained in the Starting Grant, transforming the risk models, behavioural profiles, sensing architectures and simulation environments developed in BE-SAFE into an integrated decision-support framework for cognitive and smart cities.

The new project expands both the consortium and the scope of activities. In addition to the original partners from the BE-SAFE consortium, it incorporates new expertise from Université de Pau et des Pays de l'Adour (UPPA) and Haute Ecole Spécialisée de Suisse Occidentale (HES-SO), strengthening capabilities in data integration, distributed

systems and advanced V2X communications.

BE-SAFE-Next will move the research from laboratory validation to real-world demonstration by deploying sensorised bicycle fleets and testing integrated solutions in three pilot cities: Covilhã (Portugal), Pamplona (Spain) and Braşov (Romania). Real-time data collected from these pilots will be used to calibrate AI-driven risk models, enhance cognitive simulations and support the development of proactive safety interventions aimed at reducing cyclist-vehicle conflicts in urban environments.

The long-term vision is to develop an open and transferable framework for cyclist safety that can support municipalities, transport authorities and urban planners in designing safer and more sustainable mobility systems. The consortium also intends to leverage the outcomes of BE-SAFE and BE-SAFE-Next to prepare larger international proposals, particularly under Horizon Europe and EIT Urban Mobility programmes, thereby ensuring the continuation and scaling-up of the research beyond the UNITA framework.

Beyond scientific and technological advances, the project reinforced long-term collaboration among UNITA institutions, stakeholders and local authorities, creating favourable conditions for future joint research projects, researcher mobility, educational activities, policy development and knowledge transfer initiatives focused on sustainable and safe urban mobility.

VII. FINANCIAL SUMMARY TABLE

Planned versus actual spending by item.

	Budget granted (€) UBI	Consummation (€)	% Consummation
Travel, Accommodation & subsistence	€2,000.00	€2,170.92	%109
Human resources	<i>n.a.</i>		%
Consumables	<i>n.a.</i>		%
Communication and dissemination costs	<i>n.a.</i>		%
Logistical costs / meetings / events	€500.00	€350.00	%70
TOTAL	€2,500.00	€2,520.92	%101

VIII. PLEASE ADD ONE OR TWO PHOTOS

1st In-Person Meeting (Mid-term meeting) — Braşov, Romania — March 2026



2nd In-Person Meeting (Closing meeting) — Covilhã, Portugal — September 2026





AIMAS 2026 — International Conference on Accessibility and Inland Mobility Advancement Strategies
— Sicily, Italy — April 2026



BE-SAFE at the SPARK International Partnership Meeting — UBI, Covilhã, Portugal — April 2026



BE-SAFE at the European Transport Conference 2026 — FEUP, Porto, Portugal — September 2026





BE-SAFE at the UniTBV UNITA Starting Grant dissemination event— UniTBV, Brasov, Romania — September 2026



IX. ANNEXES

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Assinado por: **BERTHA MARIA BATISTA DOS SANTOS**

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